Intro to Machine Learning
Practice Problems 3: Logistic Regression and LDA

Practice problems, not turned in

Directions: These practice problems go with Week 3 (logistic regression, including the multinomial model, and linear discriminant analysis). Problems 7 and 8 review ridge regression and the lasso from Week 2. They are not turned in, but the quizzes draw on them, so work through each one before looking at the solutions. Problems adapted from ISL cite the exercise; see the book for the original wording. Practice Problems 4 covers QDA and naive Bayes.

Notation follows the lecture. Classes are $k = 1, \dots, K$, $\pi_k = \Pr(Y = k)$ is the prior, $f_k(x)$ is the density of X in class k, and $p_k(x) = \Pr(Y = k \mid X = x)$ is the posterior. When fitting LDA by hand, use the pooled variance with divisor $n - K$.

1. **A fitted logistic model.** (ISL 4.8, Exercise 6; ISL 4.3.) Data on students in a statistics class give x_1 = hours studied, x_2 = undergraduate GPA and $Y = 1$ if the student gets an A. A logistic regression gives $\hat{\beta}_0 = -6$, $\hat{\beta}_1 = 0.05$ and $\hat{\beta}_2 = 1$.

   (a) Estimate the probability that a student who studies for 40 hours and has a GPA of 3.5 gets an A.

   (b) How many hours would that student need to study to have a 50% chance of an A?

   (c) With a 0.5 threshold, the students predicted to get an A are on one side of a line in the (hours, GPA) plane. Write the line as hours $= a + b\times$ GPA. How many hours does a student with a 3.0 GPA need?

2. **Multinomial logistic regression.** (ISL 4.3.5.)

   (a) Commuters choose to walk, bike or drive, and x is the distance to work in km. A multinomial logistic regression with *drive* as the baseline gives

   $$\hat{p}_{\text{walk}}(x) = \frac{e^{4-2x}}{D(x)}, \qquad \hat{p}_{\text{bike}}(x) = \frac{e^{2-0.5x}}{D(x)}, \qquad \hat{p}_{\text{drive}}(x) = \frac{1}{D(x)},$$

   where $D(x) = 1 + e^{4-2x} + e^{2-0.5x}$. Find the three fitted probabilities for a commuter who lives 2 km away.

   (b) At what distance are biking and driving equally likely? Walking and biking? Which mode is predicted for very large x?

   (c) How many coefficients does this model have? How many would it have with $p = 4$ predictors and $K = 5$ modes?

3. **The LDA discriminant and its boundary.** (ISL 4.8, Exercise 2; ISL 4.4.1–4.4.2, ESL 4.3.)

(a) Take $p = 1$ and $X \mid Y = k \sim N(\mu_k, \sigma^2)$, with the same σ^2 in every class. Show that the Bayes classifier assigns x to the class with the largest

$$\delta_k(x) = x\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k.$$

(b) Now take $p \geq 1$, $X \mid Y = k \sim N(\mu_k, \Sigma)$, and the lecture's $\delta_k(x) = x^T\Sigma^{-1}\mu_k - \frac{1}{2}\mu_k^T\Sigma^{-1}\mu_k + \log \pi_k$. Show that the boundary between classes k and l, $\{x : \delta_k(x) = \delta_l(x)\}$, is a hyperplane $\{x : a + x^T b = 0\}$, and give a and b.

(c) With $K = 3$ and $p = 4$, how many parameters does LDA estimate (priors, means and the shared covariance)? How many coefficients does multinomial logistic regression estimate? The LDA posterior has the same form as a multinomial logistic regression (lecture §2.2), so what is LDA doing with the extra parameters?

4. **LDA with one predictor.** (ISL 4.8, Exercise 7; ISL 4.4.2.) We want to predict whether a stock will issue a dividend this year (yes or no) from X, last year's percentage profit. Among companies that issued a dividend, X averaged $\bar{x} = 10$; among those that didn't, $\bar{x} = 0$. The variance of X in both groups is $\hat{\sigma}^2 = 36$, and 80% of companies issued dividends. Assume X is normal in each group. Find the probability that a company with $X = 4$ issues a dividend this year.

5. **Fitting LDA by hand.** (ISL 4.4.2.) A training set has one predictor:

$$\text{class 1: } x = 1, 2, 3; \qquad \text{class 2: } x = 5, 6, 6, 7.$$

(a) Compute $\hat{\pi}_k$, $\hat{\mu}_k$ and the pooled variance $\hat{\sigma}^2$.

(b) Find the LDA decision boundary. Which way did the unequal priors move it from the midpoint of the means?

6. **LDA or logistic regression?** (ISL 4.5; lecture §5; the LDA vs. logistic lab.) For each situation, say which you would try first and why.

(a) $n = 40$, $p = 3$, and histograms of each predictor look roughly normal, with about the same spread in both classes.

(b) $n = 50{,}000$, and most predictors are binary indicators or heavily skewed counts.

(c) The two classes are perfectly separated in the training data, and you need probabilities.

(d) One class contains a handful of extreme values far from the other class.

(e) True or false: since LDA and logistic regression have posteriors of the same form, fitting both to the same data gives the same coefficients.

7. **Ridge and lasso shrinkage.** (ISL 6.2.2.) In the special case of ISL 6.2.2 ($n = p$, $\mathbf{X}$ the identity matrix, no intercept), least squares gives $\hat{\beta}_j = y_j$, and ridge and the lasso have closed forms:

$$\hat{\beta}_j^R = \frac{y_j}{1+\lambda}, \qquad \hat{\beta}_j^L = \begin{cases} y_j - \lambda/2 & \text{if } y_j > \lambda/2, \\ y_j + \lambda/2 & \text{if } y_j < -\lambda/2, \\ 0 & \text{if } |y_j| \leq \lambda/2. \end{cases}$$

(a) With $y = (3, -1.5, 0.8, -0.2)$ and $\lambda = 2$, compute the ridge and lasso estimates.

(b) Which method set coefficients exactly to zero? Which shrank the largest coefficient the most?

(c) What do both estimates become as $\lambda \to 0$? As $\lambda \to \infty$?

8. **Ridge or the lasso?** (ISL 6.2.3.) For each situation, say which you would try first and why.

(a) There are 200 predictors, and you believe only a handful of them affect the response.

(b) There are 200 predictors, and you believe most of them have small effects.

(c) You need a model with only a few predictors that you can explain to a manager.

(d) True or false: before fitting ridge or the lasso, you should standardize the predictors.